Shibo Yan

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Education

Southwest University | Software Engineering | Full-time
College of Computer and Information Science & College of Software (the College)
Master's Program Sep 2024 - Jun 2027
Campus Honors: Outstanding Individual in Academic and Technological Innovation; First-Class Academic Scholarship; Outstanding Individual in Campus Cultural Activities, among others.
Bachelor's Degree Sep 2019 - Jun 2023
GPA: 3.95/5.00; rank in major: 8/124.
Course grades: Data Structures 98, Probability Theory 98, Data Mining 94, Software Modeling and Design 96.
Campus Honors: Second-Class Scholarship; Merit Student; Civic Conduct Award; Outstanding Communist Youth League Member, among others.

Research Experience

Southwest University | RISE | Safe Control & Formal Verification Sep 2024 - Jun 2027
Centre for Research and Innovation in Software Engineering | Trustworthy Software Laboratory
Research: learning-aided reach-avoid controller synthesis with formal probability bounds for nonlinear stochastic systems.

- Participate in a National Natural Science Foundation of China General Program project, "Polynomial Algebraic Methods for Safety Verification of Neural-Network-Controlled Systems" (No. 62472362; Jan 2025 - Dec 2028).
- **Conference:** **Shibo Yan**, Xia Zeng, Hengjun Zhao, Zhengfeng Yang, Zhiming Liu. *Learning-Aided Reach-Avoid Controller Synthesis with Formal Guarantees for Stochastic Systems*. MEMOCODE 2026 (ESWEEK 2026). 1st author, published.
- **Journal:** Xia Zeng, **Shibo Yan**, Zhiming Liu, Zhengfeng Yang. *Safety Guarantees for Nonlinear Systems with Bayesian Neural Network Control Policies*. *Formal Aspects of Computing* (CCF-B). 2nd author (supervisor 1st), submitted.
- **Conference:** Xia Zeng, **Shibo Yan**, Hengjun Zhao, Zhiming Liu, Zhengfeng Yang. *Formal Reach-Avoid Controller Synthesis for Stochastic Systems via Iterative Neural-Symbolic Learning*. AAAI (CCF-A, round-1 review passed). 2nd author (supervisor 1st), submitted.
- **Conference:** DAC 2027 (CCF-A). 1st author, manuscript completed, submission planned.

Projects and Internship Experience

Larac-Drive | Vehicle RL & Probabilistic Reach-Avoid Verification | Independent Developer Mar 2025 - Present

- **Background:** Developed learning-based joint longitudinal and lateral control for adaptive following, sudden lead-vehicle braking, cut-ins and local obstacle avoidance on structured roads under stochastic disturbances and motion constraints, addressing the lack of safety-boundary guarantees in unconstrained RL and degraded control smoothness in critical conditions.
- **Work:** Built a control-affine kinematic model in Frenet coordinates, with longitudinal acceleration and equivalent path curvature as inputs. Trained SAC policies and implemented IDM/MPC baselines. Designed first- and high-order control barrier functions (CBF/HOCBF) for safe longitudinal time headway and lateral road/obstacle envelopes. Enforced online safety constraints via convex quadratic programming (QP), with fallback braking when infeasible. For the closed-loop longitudinal emergency collision-avoidance subsystem under stochastic lead-vehicle braking, used PAC scenario sampling to approximate the control law with a low-degree polynomial controller. Applied variable reparameterization and SOS/SDP optimization to synthesize stochastic barrier certificates (SBCs), formally compute reach-avoid probability lower bounds and iteratively improve them using the certificates' expected-decrease condition.
- **Results:** Completed closed-loop simulations and baseline comparisons across stochastic steady-following, emergency-braking, cut-in and local obstacle-avoidance scenarios. Relative to unconstrained policies, online CBF/HOCBF-QP substantially reduced safety violation rates in extreme cut-ins and hard braking, suppressed high-frequency control oscillations, and met real-time closed-loop requirements for single-step inference and optimization. Iteratively refined polynomial controllers for the stochastic-braking emergency subsystem yielded tight reach-avoid probability lower-bound guarantees consistent with Monte Carlo statistics.

Facial Recognition Attendance System | Undergraduate Capstone Dec 2022 - Jun 2023
Second Prize, University Outstanding Undergraduate Thesis Award. Built a multi-platform attendance system using face detection and recognition on captured or uploaded images, with attendance summaries.

Chongqing Chinasoft International Information Technology Co., Ltd. Jun 2022 - Aug 2022
Internship Project Team Technical Lead & Project Manager | Outstanding Intern; score: 94.
Coordinated **MicFamily** development and integration: karaoke venue management (Spring Boot / MySQL / uni-app).

Honors and Awards

Second Prize, "Huawei Cup" China Post-Graduate Mathematical Contest in Modeling Sep 2024 - Sep 2024
Second Prize (Honourable Mention), 9th "Certificate Authority Cup" Math China International Mathematical Contest in Modeling (CAMCM) Dec 2020 - Dec 2020
Second Prize, 13th and 14th National College Student Mathematics Competition (Chongqing Division) Nov 2021 - Nov 2021

Leadership and Activities

The College, Southwest University | Research Assistant Sep 2023 - Jul 2024
TIPCC, Tsinghua University Press | Appointed Volunteer May 2022 - May 2024
National Higher-Education Expert Committee on Computing and Industry Practice Resource Development
Shapingba District Youth Talent Grassroots Service Program Sep 2021 - Aug 2022
Publicity and Planning Team Lead | University Outstanding Project, 2021 - 2022
The College: Student Union | Academic Affairs & Technology Department Member Sep 2019 - May 2020
Baidu Songguo Elite Program | Outstanding Program Graduate; ranked 2nd by total points Sep 2021 - Dec 2022